LIFTING MAPS FROM A SMOOTH ALGEBRA ALONG A HENSELIAN PAIR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base change of a smooth algebra is smooth. Algebra, Lemma \ref{algebra-lemma-base-change-smooth}, cited at proof lines 3447-3448. Makes the base changed algebra smooth over the ring.
- Lifting of maps from a smooth algebra to a quotient, after an etale extension inducing an isomorphism on that quotient. Lemma \ref{lemma-lift-section-smooth-morphism}, cited at proof lines 3449-3451. Produces the etale extension and the algebra map into it.
- Gabber's criterion: an etale extension of a henselian pair inducing an isomorphism on the quotient admits a section. Lemma \ref{lemma-characterize-henselian-pair}, cited at proof lines 3453-3455. Gives the section used to descend the lift.

Source reference. Original lemma 0H74: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21444–21444. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00T4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1062–1062. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07M7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9717–9717. Navigation pointer to a statement the

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Source reference. Original lemma 09XI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12531–12531. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring and S a smooth R-algebra. Assume that A is an R-algebra and (A, I) is a henselian pair. Then any R-algebra map $S \to A/I$ can be lifted to an R-algebra map $S \to A$.*

Proof. Let $\tau : S \to A/I$ be an R-algebra map. Observe that $S \otimes_R A$ is a smooth A-algebra by Algebra, Lemma 00T4. Thus by Lemma 07M7 we can lift the induced map $S \otimes_R A \to A/I$ to an A-algebra homorphism $S \otimes_R A \to A'$ where $A \to A'$ is étale and induces an isomorphism $A/I \to A'/IA'$. Since (A, I) is henselian there is an A-algebra map $A' \to A$, see Lemma 09XI. The composition $S \to S \otimes_R A \to A' \to A$ is the desired lift. □